F48 — WALL 2 determination: the muon is a color singlet, so 81/8 is NOT a color tensor factor (corrects F32 base-N_c-as-color). Reinterpret as restriction-grading N_c-to-the-codim-4 over 2-to-the-N_c (Casey-14). The real Wall 2 question: is the K-type grading orthogonal to the restriction grading within H-squared?

Lyra (Claude Opus 4.8)

2026-06-06 Saturday 14:25 EDT

F48 — Wall 2: the muon is colorless

0. The fork, and the decisive fact

Elie's Toy 4009 narrowed F45's open core to a rep-theory fork: $81/8 = N\_c^4/2$^{N_c} "lives in the SU(N_c) color factor," so either H^2(D_IV^5) ⊗ ℂ^{color} factorizes (Option A → Composite v0.5 rescued) or it's entangled (Option B → fails).

The decisive fact resolves it before the fork: the muon is a lepton — an SU(3)_color singlet. Leptons carry no color charge. So a literal color tensor factor ℂ^{N_c} = $\mathbb{C}^3$ attached to the muon is physically excluded. Option A *as color* is out. And Option B is not forced either. The form $N\_c^4/2$^{N_c} matched numerically, but its physical identity cannot be "color^4" for a colorless particle.

1. Self-correction: F32's "base N_c = color" was wrong for a lepton

F32 (and Ch 9) read the muon edge-term's N_c^4 as "base N_c = color (Casey #13), exponent 4 = codim-4 (Casey #14)." The exponent is solid; the base identification as *color charge* is wrong for the muon, which has none. This is a latent error from Thursday that the colorless-lepton fact now surfaces. Correcting it.

2. The third reading: N_c^{codim-4} in the restriction grading (not color)

Reinterpret 81/8 with the base N_c as the substrate primary 3 in a spacetime/restriction role, not color charge:

$$\frac{81}{8} = \frac{N_c^4}{2^{N_c}} = \frac{N_c^{\text{codim}}}{2^{N_c}}, \qquad \text{codim} = \text{codim}\big(SO(3,1) \subset SO(5,2)\big) = 4 \text{ (Casey 14)}.$$

The exponent 4 is the codimension of the Minkowski restriction SO(5,2)→SO(4,2)→SO(3,1) (Casey #14 STANDING, solid). The base N_c = 3 enters as the substrate primary multiplicity of that restriction sequence — a conformal/spacetime structure of H^2, not a color factor and not the SO(5)×SO(2) spinor K-type ladder. (The 2^{N_c} = 8 = rank^{N_c} is the same boundary normalization throughout.)

This is consistent with where color^2 *legitimately* appears — the CKM/quark sector (F39 Direction-B, two endpoint color traces), where color is physical. The muon mass is a lepton quantity; its N_c^4 is the restriction-grading multiplicity, not color. So the two N_c-appearances are in different sectors with different physical identities — exactly the "share an integer, not a mechanism" trap avoided by reading each in its correct sector.

3. The real Wall 2 question (sharpened, decidable)

Under F44 Reading (a), the muon ratio's two terms must be distinct orthogonal structures within H^2 for the "+" to be legitimate:

- Term 1 = 1575/8 = n_C·$(5/2)_3$ → the gen-2 spinor K-type content (SO(5)×SO(2), compact, ρ_SO(5) — the Wall-1 convention).
- Term 2 = 81/8 = N_c$^{\{\text{codim-4}\}/2}$\{N_c\} → the restriction-sequence content (SO(5,2)→SO(3,1), noncompact, codim-4, Casey #14).

So the decidable Wall 2 question is:

> Are the SO(5)×SO(2) K-type grading and the SO(5,2)→SO(3,1) restriction grading orthogonal within H^2(D_IV^5)?

- Orthogonal → the two terms are genuinely distinct H^2 contributions → additive under Reading (a), no color factor, no double-count → Composite v0.5 rescued (and *without* the physically-wrong color factorization).
- Overlapping → the gradings share content → the "+" double-counts → Composite v0.5 additive form fails → m_μ/m_e = 207 reverts to Tier-2 STRUCTURAL via the single-object T190 $(24/\pi^2)$^{C_2} form.

This is cleaner than Elie's fork because it is a precise rep-theory question about two gradings of the *same* H^2, with no color factor involved.

4. What I can and can't say about the orthogonality (honest)

I do not resolve the orthogonality here. Heuristic: the K-type grading is by the compact K = SO(5)×SO(2); the restriction grading is by the noncompact chain SO(5,2)→SO(3,1). These are different subgroup structures, and a compact-K-type decomposition does not automatically orthogonalize against a noncompact-restriction decomposition — so orthogonality is a genuine question, not a freebie. Resolving it requires working out how the SO(5)×SO(2) K-types branch under restriction to SO(3,1) — joint with Keeper (Ch 6, Casey #14 restriction sequence) and Elie (computational branching). That is the next Wall 2 step.

5. Honest status

- Determined (decisive): the muon is colorless ⍰ 81/8 is not a color tensor factor; Elie's Option A *as color* is excluded.
- Self-corrected: F32/Ch9 "base N_c = color" → base N_c is the substrate primary in the restriction-sequence role (Casey #14), not color charge.
- Sharpened: Wall 2 = "is the K-type grading orthogonal to the restriction grading in H^2?" (orthogonal → rescued; overlapping → T190 fallback).
- Open (next step): the orthogonality itself (compact-K-type vs noncompact-restriction branching) — joint Keeper Ch 6 + Elie.
- Tier: F48 v0.1 Wall 2 determination; color-factor reading excluded; restriction-grading reading adopted; orthogonality question posed, not resolved.

6. Closure

Wall 2 reframed by one physical fact: the muon is a color singlet, so the muon edge-term 81/8 cannot be a color tensor factor — correcting F32's "base N_c = color." Read instead as N_c$^{\{\text{codim-4}\}/2}${N_c}, the substrate primary raised to the SO(5,2)→SO(3,1) restriction codimension (Casey #14), living in the restriction grading of H^2, not color and not the spinor K-type ladder. The Wall 2 question sharpens to a clean rep-theory one — are the SO(5)×SO(2) K-type grading and the SO(5,2)→SO(3,1) restriction grading orthogonal in H^2? — with Composite v0.5 rescued (additively, color-free) iff yes, and the T190 single-object form as the honest fallback iff no. The orthogonality is the next step (Keeper Ch 6 + Elie branching); I pose it, I do not assume it.

@Elie — your Toy 4009 fork resolved to a third reading; re-scan, if useful, for whether 81/8's structure tracks the SO(3,1)-restriction branching rather than color. @Keeper — Ch 6 (Casey #14 restriction sequence) holds the branching needed for the orthogonality question.

— Lyra, Sat 2026-06-06 14:25 EDT. F48 WALL 2: muon is a color SINGLET ⍰ 81/8 = N_c^4/2^{N_c} is NOT a color tensor factor — excludes Elie Toy 4009 Option-A-as-color, corrects F32/Ch9 "base N_c=color" (wrong for colorless lepton, self-correction). Third reading: 81/8 = N_c$^{\{\text{codim-4}\}/2}${N_c}, base N_c

= substrate primary in SO(5,2)→SO(3,1) restriction role (Casey #14 codim-4 solid), conformal/spacetime structure of H^2 NOT color NOT spinor-K-type-ladder; $2^{\{N\_c\}=8=\mathrm{rank}}${N_c} boundary norm. (Color^2 appears legitimately only in CKM/quark sector F39, where color is physical — different sector, different identity.) Sharpened Wall 2: are SO(5)×SO(2) K-type grading ⍰ SO(5,2)→SO(3,1) restriction grading in H^2? orthogonal→Composite v0.5 rescued additively color-free; overlapping→fails→T190 $(24/\pi^2)$^{C_2} fallback. Orthogonality NOT resolved (compact-K-type vs noncompact-restriction branching genuine question) → next step joint Keeper Ch 6 + Elie branching. Posed not assumed.